SAYAN DUTTA

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BS-MS student in Computer Science and Data Analytics at IIT Patna with research experience in Large Language Models, NLP, and reasoning systems. Research Intern at BITS Pilani and co-author of an ACL ARR 2026 submission. Research interests include AI agents, memory-augmented systems, cognitive architectures, multilingual reasoning, and continual learning.

EDUCATION

Indian Institute of Technology Patna 2025 – 2030 (Expected)
BS-MS, Computer Science and Data Analytics (Integrated 5-Year)

- CPI: 8.64 / 10.0
- Coursework: Data Structures & Algorithms, Linear Algebra, Probability & Statistics, Optimization Fundamentals, Object-Oriented Programming, Operating Systems, Database Management Systems, Computer Networks

RESEARCH INTERESTS

Large Language Models · AI Agents · Reasoning Systems · Memory-Augmented AI · Cognitive Architectures · Multilingual NLP · Continual Learning

EXPERIENCE

Research Intern — BITS Pilani (Remote) Apr 2026 – May 2026
Project: ECOT-ERG: Empathetic Dialogue Generation with Structured Chain-of-Thought | Supervisor: Dr. Srishti Gupta, PostDoctoral Fellow, Dept. of Computer Science

- Designed and implemented ECOT-ERG, a modular reasoning framework integrating structured chain-of-thought supervision, emotion prediction, cause identification, and T5-based response generation for empathetic dialogue.
- Designed modular emotion prediction and cause identification components enabling independent ablation of individual reasoning stages without full system retraining.
- Fine-tuned T5 models using FP16 mixed-precision training and constructed DPO preference pairs from ranked responses for empathy alignment.
- Conducted 16 ablation experiments evaluating the impact of structured reasoning, emotion prediction, cause identification, and DPO alignment using BERTScore, EASE, Cause-F1, and Relevance-Accuracy.
- Performed qualitative error analysis, LLM prompting experiments, and chain-based inference to support model evaluation and the ACL ARR 2026 research manuscript.

PUBLICATIONS

Towards Causally Faithful Empathetic Dialogue Generation via Structured Appraisal Reasoning
Srishti Gupta, Ayushman Sarkar, Sayan Dutta, Shashank Vishwakarma, Rishikesh Raj, Arghya Jana, Nagandla Tarun Teja
ACL ARR 2026 (Under Review) : OpenReview: https://openreview.net/forum?id=nq94vrulUm

Contributions: Developed and trained the ECOT-ERG architecture integrating structured chain-of-thought reasoning, emotion prediction, and cause identification; constructed reasoning-chain supervision targets, conducted 16 ablation experiments, and performed qualitative and quantitative evaluation.

PROJECTS

Blix – Cognitive Agent Architecture with Online Policy Learning 2024 – Present
Independent Project | Python, PyTorch, HuggingFace Transformers, sqlite-vec, FastAPI, Pydantic
Link: github.com/SAYANDUTTA8442/blix.ai

- Designed a 30+ module cognitive architecture unifying hierarchical graph-vector memory (**HGSHM**), causal reasoning, truth maintenance, and reflective metacognition for memory-augmented AI agents — validated by 1,962 passing tests.
- Built **ADMA**, a gradient-free adaptation layer using Thompson-sampling contextual bandits to tune retrieval weighting, planning aggressiveness, and prompt style from live reward signals, with automatic rollback on confidence regression - no retraining loop required.
- Implemented a beam-search planner with global-workspace-style coordination (verifier, evaluator, specialist consensus voting) and a failure-memory/replanning loop for autonomous recovery from failed tasks.
- Developed a curiosity-driven research loop (knowledge-gap detection → hypothesis generation → experiment planning) enabling self-directed exploration without external supervision.
- Exposed the full system via a FastAPI layer (20 routers) for programmatic access to memory, planning, causal reasoning, and reflection subsystems — fully local and privacy-preserving on consumer hardware.

- Built a custom evaluation harness (**StateBench**) covering state tracking, contradiction detection, retrieval fidelity, replan success rate, and confidence calibration — used to validate subsystem behavior independently rather than relying on end-to-end testing alone.
- Designed a dependency-injected ablation framework (ablation_v3) allowing any of 7 ADMA components to be swapped or stubbed independently, enabling controlled experiments to isolate which architectural choices actually drive performance gains.
- Implemented an agent execution loop (executor, task runtime, tool-reliability tracking, tool-success prediction) alongside a document/media ingestion pipeline (PDF, DOCX, OCR) for grounding agent knowledge in external sources.

ASES - AI-Driven Precision Agriculture Advisory Platform (Capstone Project) Jan 2026 – May 2026
Python, Scikit-learn, Google Gemini API, Streamlit, SQLite
Live Link: sayanduttaiitpatnacda107.streamlit.app/

- Developed crop and fertilizer recommendation modules using feature-normalized KNN models over multi-dimensional soil and environmental attributes.
- Enhanced recommendation quality by integrating LLM-based contextual advisory generation through structured prompting with Google Gemini API.
- Designed a lightweight end-to-end application with Streamlit frontend and SQLite backend for persistent storage and local deployment.
- Established a modular pipeline separating prediction, recommendation, and natural-language advisory components for extensible agricultural decision support.

TECHNICAL SKILLS

Languages: Python, C++, C, Java
ML / DL: PyTorch, Hugging Face Transformers, Scikit-learn, Seq2Seq Training
LLM: Chain-of-Thought (CoT), Direct Preference Optimization (DPO), Retrieval-Augmented Generation (RAG), Prompt Engineering
NLP: Sentence Transformers, T5, Empathetic Dialogue Systems, BERTScore, EASE, Cause-F1
AI Systems: FAISS, FastAPI, REST APIs, Pydantic, sqlite-vec
Tools & Platforms: Git, Linux, CUDA, FP16 Mixed Precision, Jupyter Notebook, Docker, Google Cloud
Databases: MySQL, SQLite

CERTIFICATIONS

Google Cloud Skills Boost — 70 completions (Apr–Jun 2025), including Introduction to Large Language Models, Generative AI, and Responsible AI; Gemini-based GenAI application development; MLOps for AI agents. Full profile: https://www.skills.google/public_profiles/7134a42a-d7bf-453e-ab38-0570038bf2a8

LEADERSHIP

Lead Google Student Ambassador 2025 – Present
Indian Institute of Technology Patna

- Coordinated technical outreach initiatives serving 100+ students across AI/ML workshops and developer events.

ACHIEVEMENTS

- Co-author of an ACL ARR 2026 submission on structured chain-of-thought reasoning for empathetic dialogue generation.
- Selected as a Research Intern under Dr. Srishti Gupta, Department of Computer Science, BITS Pilani.
- Lead Google Student Ambassador at IIT Patna, organizing AI/ML workshops and technical outreach initiatives for 100+ students.
- Conducted 16 ablation experiments and contributed to DPO alignment, error analysis, and evaluation pipelines for ECOT-ERG.
- Developing Blix, an independent research platform for memory-augmented AI agents, cognitive architectures, planning, and reasoning.